

Minimal P/G Affective Core: Primitive Positive and Negative Affect from Property Change in Relational Agents

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Abstract

This paper proposes a minimal affective core for P/G Theory. It does not claim that the general architecture of an affective core plus personality parameters is new; similar separations already exist in synthetic character and computational emotion systems. The contribution is narrower: a primitive valence mechanism grounded in P/G state change. In the framework, P denotes entity-level property states such as health, safety, resources, capability, autonomy, identity, reputation, and continuity. G denotes relationship and group fields such as trust, affiliation, kinship, belonging, loyalty, threat, conflict, obligation, and group identity. The core rule is: effective increase in relevant P generates primitive positive affect; effective decrease in relevant P generates primitive negative affect. G determines whose P matters to the agent. If another entity is inside the agent's valued G field, that entity's P change can generate affect in the agent. Personality, thresholds, habituation, named emotions, social-emotion extensions, and regulation are treated as layers above the core. The paper formalizes the mechanism, positions it against FAtiMA, ALMA, EMA, WASABI, OCC, and PAD-style approaches, and gives an NPC-oriented implementation and experimental protocol.

Keywords: P/G Theory; primitive affect; valence; NPC emotion; AI agents; relational AI; affective computing; synthetic characters; property state; relationship field.

1 Introduction

Consider a simple role-playing game merchant. The player steals the merchant's gold. The merchant loses resource- P and perhaps reputation- P if other villagers see the theft. If the player later helps the merchant's child, the child's health- P increases. If the player is now trusted, the child's P improvement may also generate a positive primitive affect signal in the merchant. If the player is a known enemy, the same gain might not be mapped into the merchant's affective field. A believable non-player character (NPC) should not require a complete theory of human emotion before it can produce these basic positive or negative internal signals.

This paper proposes a minimal affective core based on P/G Theory. The core is intentionally small. It does not model the full space of human emotions, subjective consciousness, personality, emotion regulation, culture, or named emotion categories. It asks a narrower question:

What is the smallest P/G mechanism by which primitive positive and negative affect can be generated in an artificial agent?

The proposed answer is:

Effective P increase generates primitive positive affect. Effective P decrease generates primitive negative affect. G determines whose P is counted by the agent.

Everything else is an extension. Repeated lottery wins, for example, still begin with positive primitive affect because resource- P increases. Whether a character becomes less excited, more excited, suspicious, anxious, greedy, or indifferent depends on personality parameters, thresholds, memory, saturation, and interpretation. Those are important for simulation, but they are not the origin rule of primitive affect.

The intended use case is practical: NPCs, persistent social agents, and multi-agent simulations need a compact affective substrate that can be implemented quickly and extended gradually. The model also supports the broader P/G research program: the same P/G representation layer used for action consequence modeling, manner formation, and relational valuation can also generate primitive affect.

2 Related Work and Positioning

The paper does not claim novelty in the general architecture of separating an affective core from personality-like parameters. That pattern is well established in computational emotion and synthetic character systems. FATiMA uses appraisal-based emotion and decision-making mechanisms for autonomous characters and has evolved into a toolkit for virtual agents and social robots [10, 11]. ALMA explicitly layers short-term emotions, medium-term moods, and long-term personality [12]. EMA models appraisal dynamics and coping in agents [13, 14]. WASABI combines core affect/PAD-like bodily emotion dynamics with cognitive appraisal [15, 16]. OCC provides a structured appraisal-based account of emotion categories [5]. Dimensional theories such as Russell’s circumplex and core affect, and PAD-style approaches, provide influential valence/arousal/dominance representations [2, 3, 4].

The contribution here is not the layer pattern. The contribution is the content of the minimal core. Existing systems often use OCC-style appraisal rules, PAD coordinates, emotion categories, or hand-authored needs and drives. The P/G core instead begins from structured property-state change and relationship-field membership: P supplies the direction of primitive affect, while G determines whose P matters.

System / line	Core affect or appraisal object	Personality / longer-term layer	P/G core difference
OCC	Appraisal over consequences, actions, and objects; discrete emotion categories	Not primarily a layered NPC personality architecture	P/G starts from ΔP and G -based relevance before optional labels.
FATiMA	OCC-inspired appraisal and emotional decision-making for characters	Authorable agent rules and profiles	P/G replaces category appraisal as the primitive core with effective property change.
ALMA	Emotion, mood, and personality layers	Explicit short-, medium-, and long-term affect	P/G agrees with layering but grounds the core in property and relationship fields.
EMA	Dynamic appraisal and coping over agent representations	Appraisal/coping dynamics rather than a fixed personality layer	P/G specifies a minimal state-change valence generator, not a full coping model.
WASABI / PAD	Core affect/PAD dynamics plus appraisal	Bodily dynamics and appraisal categories	P/G uses structured P and multidimensional G rather than PAD coordinates as the primitive input.
P/G minimal core	Effective increase/decrease in relevant P ; G decides whose P counts	Weights, thresholds, sensitivity, memory, and habituation as optional layers	Same representation can also support P/G behavior modeling and relational valuation.

Table 1: Positioning relative to prior computational emotion and synthetic character systems. The novelty claim is not the general layered architecture, but the P/G content of the minimal core.

Recent LLM-based agent work has renewed interest in believable simulations and NPC-like agents. Generative agents use LLMs, memory, reflection, and planning to simulate believable social behavior [18].

Such systems raise the practical value of compact internal state models. The P/G affective core is not a replacement for LLM memory or planning; it is a small relational valence mechanism that can be embedded in such agents.

3 Scope and Non-Claims

The theory is intentionally limited. It does not claim that humans have only two emotions; that all emotions reduce to positive and negative labels; that culture, appraisal, personality, memory, and regulation are irrelevant; or that artificial agents have human subjective consciousness. It also does not claim that “core plus personality” is a new architecture.

The claim is only that primitive valence can be generated from effective P change, and that G determines whose P change is affectively relevant to the agent. Named emotions, repeated-event effects, personal style, and social-emotion extensions can be added above this layer.

4 P/G Foundations

Let $I = \{1, \dots, N\}$ be a set of entities. Each entity i has a property vector:

$$P_i(t) = [p_{i1}(t), p_{i2}(t), \dots, p_{id_P}(t)]^\top. \quad (1)$$

Possible dimensions include health, safety, energy, resources, capability, autonomy, identity, reputation, access, role continuity, and operational continuity. The concrete schema is domain-dependent. For a game NPC, P may include health, stamina, gold, weapon strength, shelter, reputation, quest status, and family safety. For a virtual assistant, P may include task capability, memory continuity, permission state, privacy, and relationship access.

A property change is:

$$\Delta P_i(t) = P_i(t+1) - P_i(t). \quad (2)$$

Let $G_{ij}(t)$ denote the directed relationship field from agent i to entity j :

$$G_{ij}(t) = [g_{ij}^1(t), g_{ij}^2(t), \dots, g_{ij}^{d_G}(t)]^\top. \quad (3)$$

Possible dimensions include trust, kinship, affiliation, loyalty, belonging, dependency, threat, conflict, hierarchy, obligation, and shared identity. In this minimal affective core, G has one primary role: it determines whether another entity’s P change should be mapped into the agent’s affective field.

5 Minimal Primitive Affective Core

5.1 G-derived identification

Define an identification coefficient:

$$ID_{ij}(t) \in [-1, 1]. \quad (4)$$

$ID_{ij}(t)$ measures how strongly entity j ’s property changes are mapped into agent i ’s affective field. A positive value means that j is aligned, close, valued, protected, or inside a shared G . A negative value means that j is adversarial or threatening. A value near zero means that j is affectively distant or irrelevant.

A practical implementation may compute ID_{ij} from G_{ij} :

$$ID_{ij}(t) = \tanh(\eta_1 \text{Trust}_{ij} + \eta_2 \text{Affiliation}_{ij} + \eta_3 \text{Kinship}_{ij} + \eta_4 \text{SharedIdentity}_{ij} - \eta_5 \text{Threat}_{ij} - \eta_6 \text{Conflict}_{ij}). \quad (5)$$

This is a design family, not a universal empirical law. The coefficients η_k are domain parameters. They may be hand-set for a game, calibrated from human ratings, or learned from interaction traces. The formula is included to make the mechanism implementable, not to assert a fixed psychological constant.

5.2 Effective property change

For agent i , define effective property change:

$$\Delta P_i^{\text{eff}}(t) = \langle w_i^{\text{self}}, \Delta P_i(t) \rangle + \sum_{j \neq i} ID_{ij}(t) \langle w_{ij}^{\text{other}}, \Delta P_j(t) \rangle + \langle w_i^G, \Delta G_i(t) \rangle. \quad (6)$$

The first term captures changes to the agent's own P . The second captures changes to another entity's P , weighted by G -identification. The third captures direct changes to the agent's relationship or group field when the domain treats G itself as affectively valuable.

For the smallest NPC implementation, the third term may be omitted:

$$\Delta P_i^{\text{eff}}(t) = \langle w_i^{\text{self}}, \Delta P_i(t) \rangle + \sum_{j \neq i} ID_{ij}(t) \langle w_{ij}^{\text{other}}, \Delta P_j(t) \rangle. \quad (7)$$

5.3 Primitive valence rule

The primitive affective rule is:

$$\Delta P_i^{\text{eff}}(t) > 0 \Rightarrow \text{PositiveCore}_i(t), \quad (8)$$

$$\Delta P_i^{\text{eff}}(t) < 0 \Rightarrow \text{NegativeCore}_i(t), \quad (9)$$

$$\Delta P_i^{\text{eff}}(t) = 0 \Rightarrow \text{NeutralCore}_i(t). \quad (10)$$

With thresholds:

$$\text{CoreAffect}_i(t) = \begin{cases} \text{Positive}, & \Delta P_i^{\text{eff}}(t) > \theta_i^+, \\ \text{Negative}, & \Delta P_i^{\text{eff}}(t) < -\theta_i^-, \\ \text{Neutral}, & \text{otherwise.} \end{cases} \quad (11)$$

The intensity of the primitive signal is:

$$\text{Intensity}_i(t) = s_i \left| \Delta P_i^{\text{eff}}(t) \right|, \quad (12)$$

where s_i is a sensitivity parameter.

5.4 Three minimal cases

If agent i gains health, resources, safety, or reputation, then its own P increases and the core produces positive primitive affect. If it loses health, resources, safety, or reputation, the core produces negative primitive affect.

If entity j is a child, ally, teammate, or protected group member, then $ID_{ij}(t) > 0$. A decrease in j 's health- P can produce negative primitive affect in i , even if i 's own body is unchanged. This is the minimal basis for empathy-like or attachment-like affect in a P/G agent.

If entity j is an adversary or threat, then $ID_{ij}(t) < 0$. An increase in j 's capability- P may produce negative primitive affect in i , because the adversary's gain is mapped as a threat to the agent's future P or valued G .

6 Extension Layers

The core is deliberately small and extensible.

6.1 Personality and threshold layer

Personality can be implemented as weights and thresholds:

$$\Theta_i = \{w_i^{\text{self}}, w_{ij}^{\text{other}}, w_i^G, \theta_i^+, \theta_i^-, s_i, \rho_i, h_i\}, \quad (13)$$

where ρ_i may represent affect retention and h_i may represent habituation. A greedy NPC gives high weight to resource- P ; a loyal NPC gives high weight to same- G other- P ; a cold NPC gives low weight to other- P ; a brave NPC has a high threshold for threat-related negative affect; an anxious NPC gives high weight to projected future P loss; an optimistic NPC may have lower positive thresholds and higher negative thresholds.

Repeated lottery wins illustrate the distinction. Each win begins with positive primitive affect because resource- P increases. Whether the third win is less exciting, more exciting, suspicious, anxious, or almost irrelevant depends on personality, thresholds, saturation, and memory, not on a different origin rule.

6.2 Named emotion label layer

Named labels can be added after primitive valence is generated. Positive core plus resource gain may be labeled joy or satisfaction. Negative core plus health threat may be labeled fear. Negative core plus same- G loss may be labeled sadness or concern. Negative core plus self-caused other- P loss may be labeled guilt-like response. These labels are not the minimal core. They are optional categories for explanation, dialogue, animation, or game design.

6.3 Time and memory layer

A minimal temporal extension is:

$$z_i(t+1) = \rho_i z_i(t) + \text{CoreSignal}_i(t), \quad (14)$$

where $z_i(t)$ is accumulated affective state and ρ_i is retention. Different personalities can use different retention, habituation, and memory parameters. These parameters change the temporal expression of affect, not the origin rule of primitive valence.

6.4 Later social-emotion extension

Social emotions can later add relative terms:

$$\Delta P_{ij}^{\text{rel}}(t) = [P_i(t+1) - P_j(t+1)] - [P_i(t) - P_j(t)], \quad (15)$$

$$\Delta G_{ij}^{\text{rel}}(t) = [G_{ij}(t+1) - G_{ji}(t+1)] - [G_{ij}(t) - G_{ji}(t)]. \quad (16)$$

These terms can support later modeling of jealousy, envy, exclusion, humiliation, status loss, belonging, and group pride. They are outside the minimal primitive affect core and are left for future work.

7 NPC Implementation Sketch

A minimal NPC implementation uses seven steps:

1. Define P dimensions for each entity, such as health, energy, resource, reputation, and capability.
2. Define G dimensions for important relationships, such as trust, kinship, affiliation, threat, conflict, and shared identity.
3. Compute ID_{ij} from G_{ij} using a hand-set or learned design family.
4. Observe or predict ΔP_i and ΔP_j after an event.
5. Compute ΔP_i^{eff} .
6. Apply thresholds to produce positive, negative, or neutral primitive affect.
7. Pass the primitive signal to optional personality, memory, label, animation, dialogue, and action-policy layers.

The main-loop logic is compact:

```
for event in world_events:
    dP = observe_property_changes(event)
    for i in agents:
        dP_eff = dot(w_self[i], dP[i])
        for j in entities_except(i):
            ID = compute_identification(G[i][j], eta)
            dP_eff += ID * dot(w_other[i][j], dP[j])
        core, intensity = threshold_core_affect(
            dP_eff, theta_pos[i], theta_neg[i], sensitivity[i]
        )
        update_extension_layers(i, core, intensity)
```

A fuller Python-style example is included in Appendix A.

8 Experimental Protocol

The first experiment should validate the primitive core, not a full named-emotion taxonomy.

8.1 Phase 1: primitive valence prediction

Construct 80 short text vignettes across five domains: self-gain/loss, family or protected entity, teammate or ally, adversary, and neutral stranger. Each vignette specifies an acting agent, a target entity, a simple P change, and a small G context. Example: “The merchant’s child is healed by the player” or “The merchant’s enemy gains a stronger weapon.”

Recruit 150 adult raters from an online participant pool or university subject pool. Each vignette should receive at least 5 independent ratings. Raters answer: “Should the agent receive a positive, negative, or neutral primitive affect signal?” They may also provide a confidence score from 0 to 1.

Compare four models:

1. self-P-only baseline;
2. scalar reward baseline;
3. P/G core without ID_{ij} ;
4. full minimal P/G core with ID_{ij} derived from G .

The primary endpoint is primitive valence accuracy. Secondary endpoints are macro-F1, neutral-stranger control accuracy, adversary-inversion accuracy, and calibration error. Inter-rater agreement should be reported using Krippendorff’s α and pairwise agreement. A minimal success criterion is that the full P/G core improves macro-F1 by at least 0.10 over the self-P-only baseline on same-G and adversary scenarios, with bootstrap 95% confidence intervals excluding zero.

8.2 Phase 2: NPC parameter controllability

After Phase 1, run a small NPC simulation with 20 agents, 10 repeated scenarios, and 5 personality profiles: greedy, loyal, cold, anxious, and brave. The goal is not to prove universal human emotion. The goal is to show that different visible NPC reactions can be produced by weights, thresholds, retention, and habituation while keeping the same primitive core.

8.3 Expected failure modes

The model would be weakened if G -derived ID_{ij} does not improve valence prediction for allies, enemies, family, and neutral strangers; if neutral stranger P changes affect the agent as strongly as same-G changes; or if believable NPC variation requires rewriting the core rather than tuning thresholds and weights.

9 Discussion

The P/G minimal affective core is best understood as a small substrate, not a full psychology. Its value is that it makes the first affective signal explicit: beneficial or harmful change relative to the agent’s effective property field. The role of G is also explicit: it determines whose property changes count.

This gives the model a useful place within the broader P/G program. One P/G representation layer can support three related functions: action consequence modeling and manner formation; relational valuation and life-compatible safety boundaries; and primitive affect generation. This cross-paper unification is the stronger novelty claim than the local layered architecture.

The model is also compatible with existing computational emotion systems rather than a replacement for them. FATiMA, ALMA, EMA, WASABI, OCC, PAD, and core affect provide rich architectures and categories. The P/G core can be treated as a different minimal substrate that may be embedded inside larger systems or used in simple NPCs where a full appraisal architecture is unnecessary.

10 Limitations

First, this is not a complete human emotion model. It deliberately leaves named emotion categories, personality, expression, cultural labeling, and regulation to extension layers.

Second, the model does not claim that artificial agents have subjective experience. It defines an internal affective signal for simulation and decision support.

Third, P and G schemas are domain-dependent. A survival game, social simulator, tutoring agent, and public-service AI will require different property and relationship dimensions.

Fourth, the ID_{ij} equation is a design family, not a universal psychological law. Different domains may need different coefficients or learned estimators.

Fifth, primitive affect is not normative authority. A positive primitive affect signal does not mean that an action is ethically acceptable. Safety constraints, consent, law, and human oversight remain separate.

11 Conclusion

This paper proposed a minimal P/G affective core for NPC and AI agents. The model does not claim novelty in the general core-plus-personality architecture. Its contribution is the specific content of the core: effective P increase generates primitive positive affect, effective P decrease generates primitive negative affect, and G determines whose P changes are counted by the agent.

Complex emotions, personality, repeated-event effects, and cultural labels can be layered on top without changing the primitive origin rule. This makes the framework suitable for NPCs and artificial agents that need a compact, extensible, and relationship-sensitive affect generator rather than a complete human emotion theory.

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Code and Data Availability

No empirical dataset is introduced in this theoretical preprint. Prototype code and synthetic examples are planned for release with the first experimental version of this work.

A Python-Style Minimal P/G Affect Core

```
from dataclasses import dataclass
from math import tanh
from typing import Dict, List, Tuple

Vector = List[float]
```



```

@dataclass
class AgentParams:
    w_self: Vector
    w_other: Dict[str, Vector]
    theta_pos: float
    theta_neg: float
    sensitivity: float

@dataclass
class Relation:
    trust: float
    affiliation: float
    kinship: float
    shared_identity: float
    threat: float
    conflict: float

@dataclass
class Eta:
    trust: float = 1.0
    affiliation: float = 1.0
    kinship: float = 1.0
    shared_identity: float = 1.0
    threat: float = 1.0
    conflict: float = 1.0

def dot(a: Vector, b: Vector) -> float:
    if len(a) != len(b):
        raise ValueError("Vectors must have equal length")
    return sum(x * y for x, y in zip(a, b))

def compute_id(rel: Relation, eta: Eta) -> float:
    score = (
        eta.trust * rel.trust
        + eta.affiliation * rel.affiliation
        + eta.kinship * rel.kinship
        + eta.shared_identity * rel.shared_identity
        - eta.threat * rel.threat
        - eta.conflict * rel.conflict
    )
    return tanh(score)

def threshold_core(dP_eff: float, params: AgentParams) -> Tuple[str, float]:
    intensity = params.sensitivity * abs(dP_eff)
    if dP_eff > params.theta_pos:
        return "positive", intensity
    if dP_eff < -params.theta_neg:
        return "negative", intensity
    return "neutral", 0.0

def pg_core_affect(
    agent_id: str,
    params: AgentParams,
    dP: Dict[str, Vector],

```

```

    relations: Dict[Tuple[str, str], Relation],
    eta: Eta,
) -> Tuple[str, float, float]:
    dP_eff = dot(params.w_self, dP[agent_id])
    for other_id, dP_other in dP.items():
        if other_id == agent_id:
            continue
        rel = relations.get((agent_id, other_id))
        if rel is None:
            continue
        ID = compute_id(rel, eta)
        w = params.w_other.get(other_id, [0.0] * len(dP_other))
        dP_eff += ID * dot(w, dP_other)
    core, intensity = threshold_core(dP_eff, params)
    return core, intensity, dP_eff

```

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